MAT 504

Homework 3 – From an MD Trajectory to a Defensible Diffusion Coefficient

Computer Modelling in Materials Science

Regular deadline:	See SUCourse	**Late submission:**	−20 points per late day
Estimated working time:	about 60 minutes	**Total:**	100 points

Purpose. Week 3 connected atomic trajectories to diffusion. Week 4 added the practical question of whether a trajectory and the observable obtained from it can be trusted. In this homework you will use two independent molecular-dynamics replicas to examine trajectory preprocessing under periodic boundaries, calculate the mean-squared displacement (MSD), obtain a diffusion coefficient, and use replica agreement as a simple convergence diagnostic. The central sequence is: **equilibrate, generate production data, preprocess the trajectory, calculate MSD, fit the diffusive regime, compare replicas, and make a scientific judgment.**

Submission

Submit **one completed Jupyter notebook** containing all requested figures, numerical results, and concise written answers. The notebook should run from beginning to end without manual intervention. Use appropriate axis labels and reduced units.

Assessment of computational work. You are responsible for all code and analysis in your submission, including supplied helper functions. You may be asked in person to explain any part of the notebook, including individual lines of code, equations, figures, numerical procedures, or interpretations. Your ability to explain the submitted work is part of the assessment.

Computational setup

We use the same two-dimensional Lennard–Jones fluid as in the Week 4 hands-on session, with

$$\epsilon = \sigma = m = k_B = 1.$$

The low-level MD engine is supplied. Each independent replica is prepared using **NVT equilibration** at target temperature $T = 1.0$, followed by an **NVE production trajectory** for the diffusion analysis. The helper returns **unwrapped coordinates**; wrapped coordinates can be reconstructed from the primary simulation box.

Why NVE production? A thermostat is useful for preparing the desired thermal state, but thermostat algorithms can alter real dynamics. In this exercise we therefore use the thermostat during equilibration and measure the dynamical observable D from conservative NVE production data.

Part 1 – Independent production trajectories 15 points

Generate two independent replicas using the supplied workflow with seeds `504` and `509`. All other parameters must remain identical.

1.1 Prediction and replica logic – 5 pts

Before running, answer briefly:

a. Why are different initial velocities / random seeds useful when checking convergence?

b. Should the two trajectories be identical? Should estimates of the same equilibrium transport property be reasonably consistent if sampling is adequate?

1.2 NVE energy diagnostic – 10 pts

Plot total energy per particle versus production time for both replicas. Comment on whether either production trajectory shows obvious systematic energy drift, and explain why this check should be made before analyzing diffusion.

Part 2 – Wrapped versus unwrapped motion 25 points

Periodic boundary conditions make a particle that leaves one side of the primary box appear again on the opposite side. This is convenient for simulation, but displacement-based observables require special care.

For replica 1:

1. construct a wrapped trajectory from the supplied unwrapped coordinates;
2. calculate the time-origin-averaged MSD from both wrapped and unwrapped coordinates using `max_lag=300`;
3. plot both curves on the same graph.

Answer briefly:

a. Why is an MSD calculated directly from wrapped coordinates incorrect for long-time diffusion analysis?

b. Which coordinate representation should be used to preserve physical translational displacement?

Part 3 – Obtain the diffusion coefficient 35 points

For normal diffusion in d spatial dimensions,

$$\mathrm{MSD}(t) = 2dDt + C.$$

For this two-dimensional system,

$$D = \frac{1}{4}\frac{d\,\mathrm{MSD}}{dt}.$$

For **each replica**, calculate the unwrapped MSD and fit the supplied interval

$$2 \le t \le 8$$

using a straight line. Report D_1 and D_2. Plot both MSD curves and overlay the fitted lines within the fitting interval.

Answer briefly:

a. Why is D obtained from the slope rather than from a single displacement value?

b. Why is a specified fitting interval useful in this introductory homework?

c. Why might very short or very long lag times be unsuitable for estimating a long-time diffusion coefficient?

Part 4 – Replica agreement and convergence judgment 20 points

Calculate

$$D_{\text{mean}} = \frac{D_1 + D_2}{2}$$

and

$$\%\,\text{difference} = 100\frac{|D_1 - D_2|}{D_{\text{mean}}}.$$

Answer:

a. Do the replicas give reasonably similar values of D? What does this agreement tell you, and what does it **not** prove?

b. If temperature appeared perfectly stable but the two replicas gave very different values of D, would you call D converged? Explain.

c. If the scientific objective were instead to obtain the equilibrium density at specified temperature and pressure, which ensemble would be more appropriate: NVE, NVT, or NPT? Explain in one sentence.

Final synthesis 5 points

In **4–5 sentences**, explain why a trustworthy diffusion coefficient requires more than simply running an MD simulation and plotting an MSD curve. Your answer should mention:

- the simulation protocol / ensemble;
- trajectory preprocessing under periodic boundaries;
- the linear diffusive regime;
- independent replicas or convergence.

Before you submit

Your notebook should contain:

1. two independent production trajectories and the NVE energy diagnostic;
2. the wrapped-versus-unwrapped MSD comparison;
3. two unwrapped MSD curves with fitted lines;
4. D_1, D_2, D_{mean}, and the replica percent difference;
5. all written answers and the final synthesis.